

MPC Lab – HMC Meeting

Eric Choi, Mark Pustilnik, Hansung
Kim, and Francesco Borrelli

8/25/2025

Agenda

1. Recap

2. Technical Updates

- Sample data analysis
- Suggest An energy Model to fit the data

3. Discussion

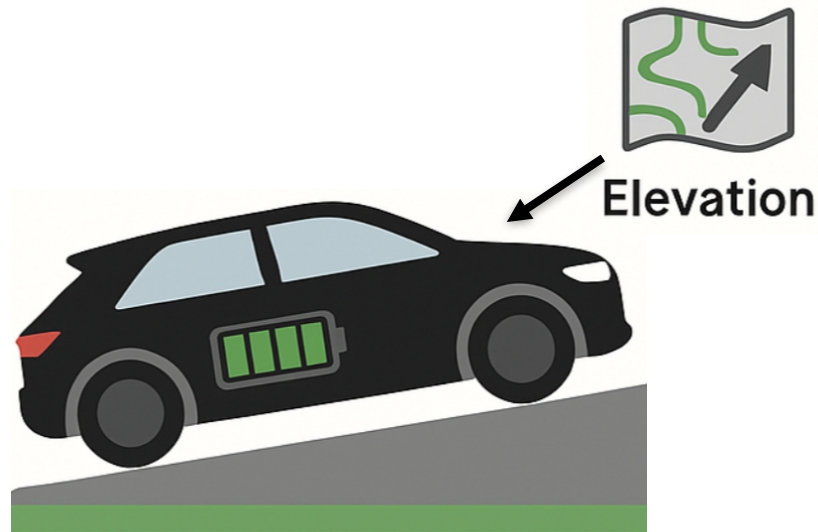
- Future Roles and Responsibilities

Recap

Project Overview

Objective: Develop and validate a road grade-aware energy consumption estimation model for electric vehicles

Duration: 7 months (June – December 2025)



UCB Vehicle Platform



Model: 2023 Hyundai IONIQ 5

Signals:

- Vehicle
 - Speed, wheel torque, etc.
- Energy
 - High/Low voltage battery power, SOC, etc.
- IMU
 - Accel., pitch angles, etc.
- GPS
 - Latitude, Longitude, and Altitude
 - (optional) Altitude through Google or HERE API

Literature Review

Academic Papers

- There are many academic papers that try to model the effects of sloped road on EV range performance
- Most give a general model and not a vehicle specific (based on mass and road grade)
- Some standout papers:
 - Oriol Travesset-Baro, Transport energy consumption in mountainous roads. A comparative case study for internal combustion engines and electric vehicles in Andorra (2015) – Talk about the difference between powertrain types on range performance on hilly road
 - Xiaodan Xu, A modal-based approach for estimating electric vehicle energy consumption in transportation networks (2019) - statistically-based energy model derived from simulator
 - Heran Shen, An Experimental Comparison of Physics-based and Machine-Learning-based Electric Vehicle Energy Consumption Estimation Methods (2023) – suggests that ML methods superior in accuracy but only on roads with enough data

Technical Update

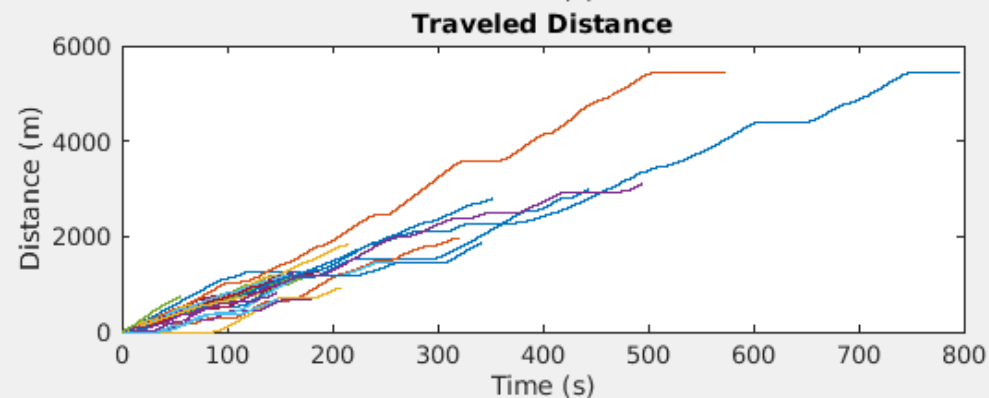
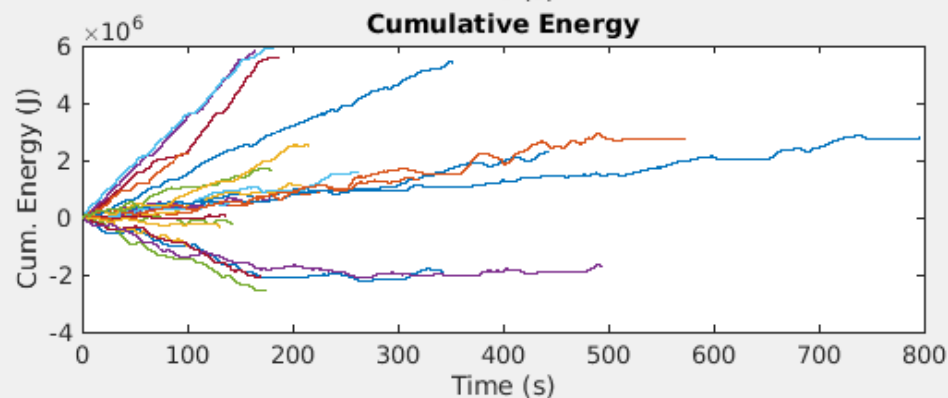
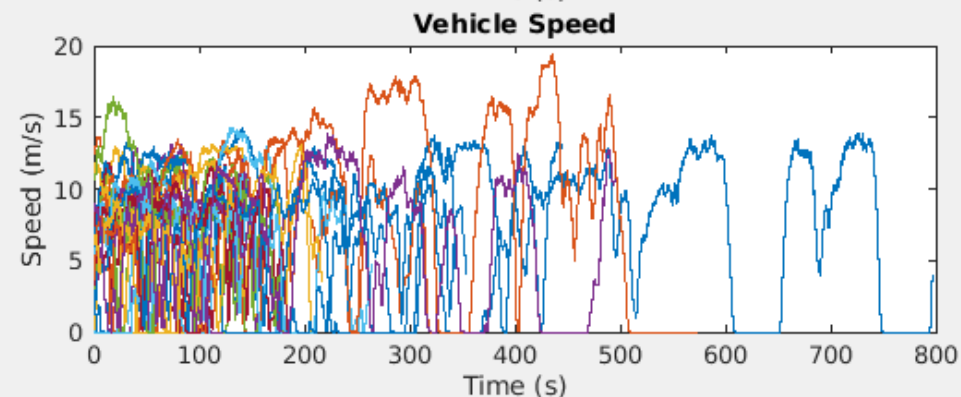
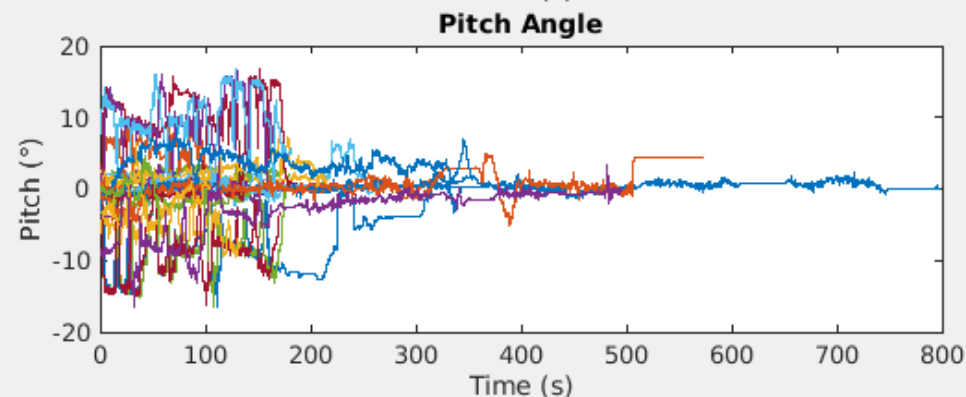
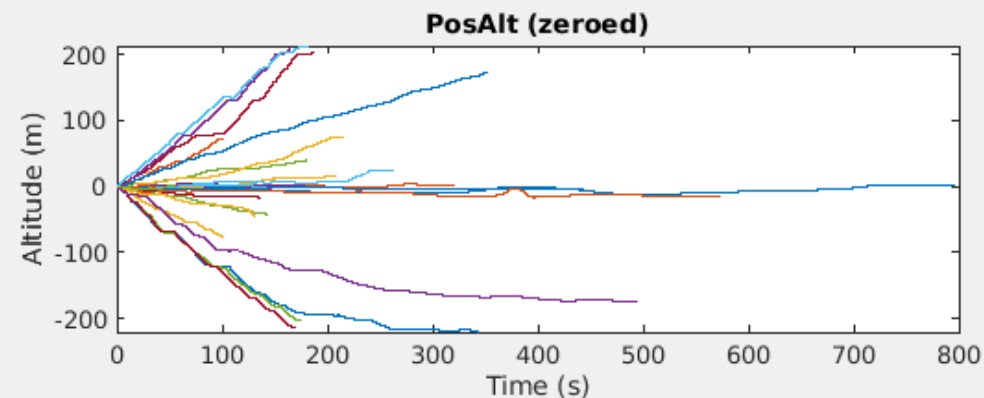
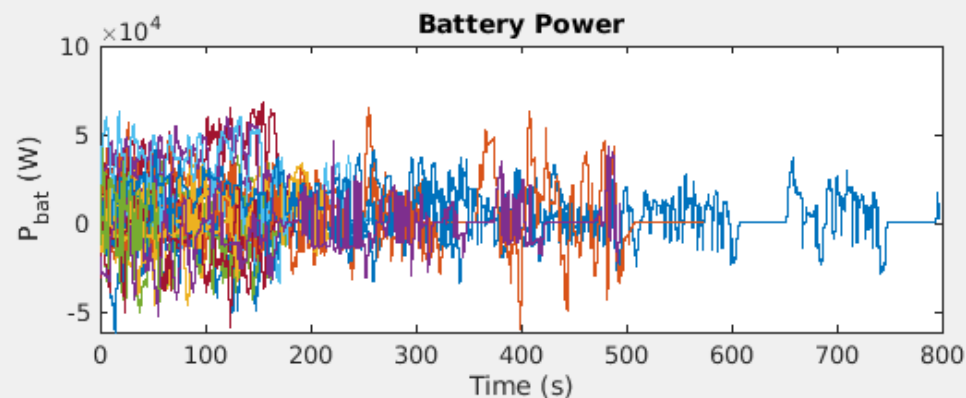
The map displays a route through Berkeley, CA, starting from San Pablo Avenue and heading towards the University of California, Berkeley. The route is marked with a blue line and includes several stops and landmarks. A callout box indicates a travel time of 35 minutes for 9.8 miles. The route passes through areas like Westbrae, Poets Corner, and Berkeley, and includes stops at various markets, parks, and educational institutions. Key landmarks include Tilden Park, Regional Parks Botanic Garden, and the University of California, Berkeley. The map also shows major roads like Highway 580 and Highway 80, and various landmarks like Tilden Park, Regional Parks Botanic Garden, and the University of California, Berkeley.

Moderate Road Grade

- Total traveled distance: ~1000 m
- Normalize all Recordings to 1000m travel distance

- Normal Driving Mode
- Regen Level 1

Sample Data Collection



Data Analysis

- We look at the net battery power associated with driving the car – total battery energy consumption minus the AC power consumption:

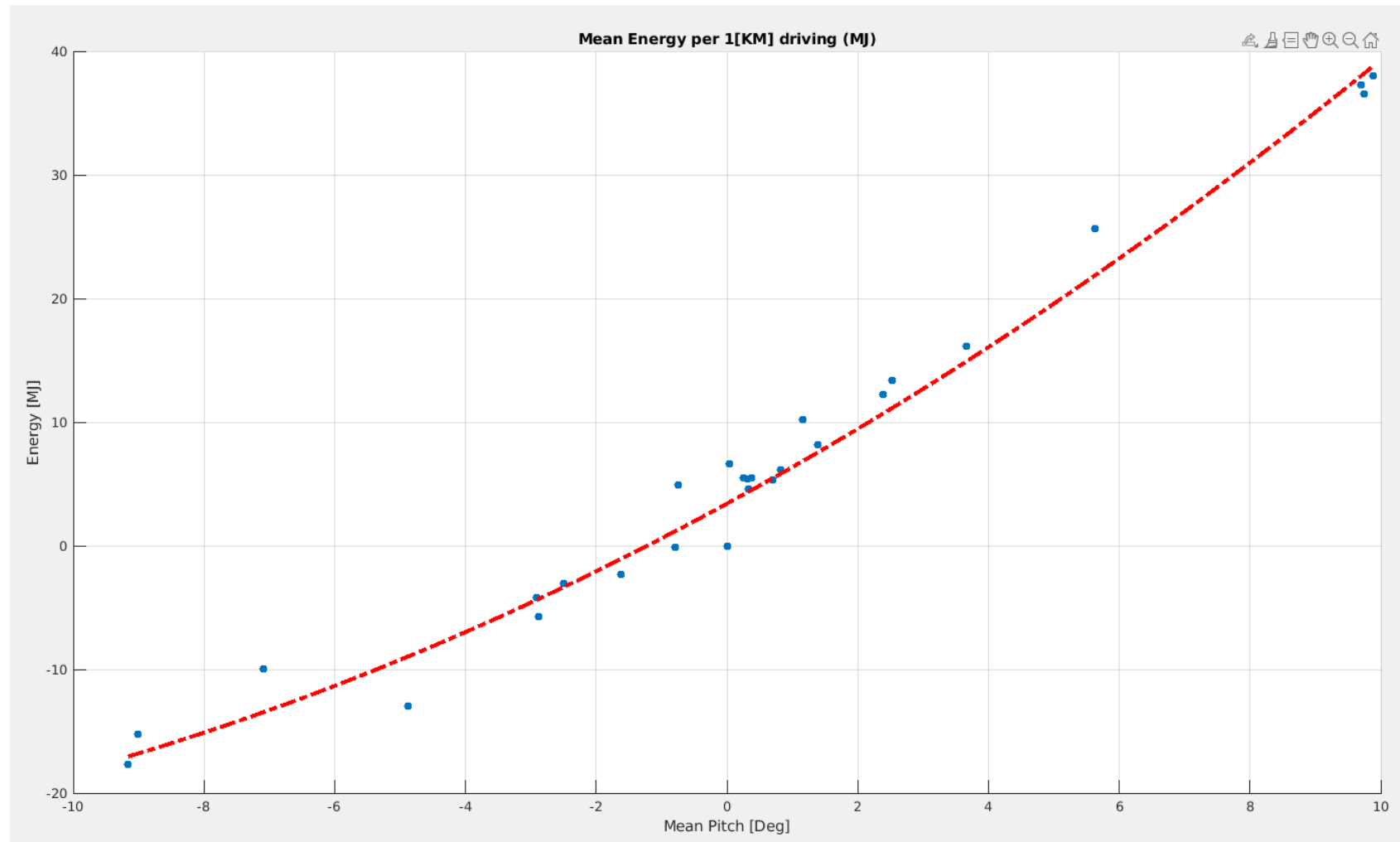
$$P(t) = V_{bat}(t) \cdot I_{bat}(t) - P_{AC}(t)$$

- The energy of each section is normalized to 1[KM]:

$$W_{tot}^{1KM} = \frac{\int_{t=0}^T P(t) dt}{\int_{t=0}^T V(t) dt} \cdot 1[KM]$$

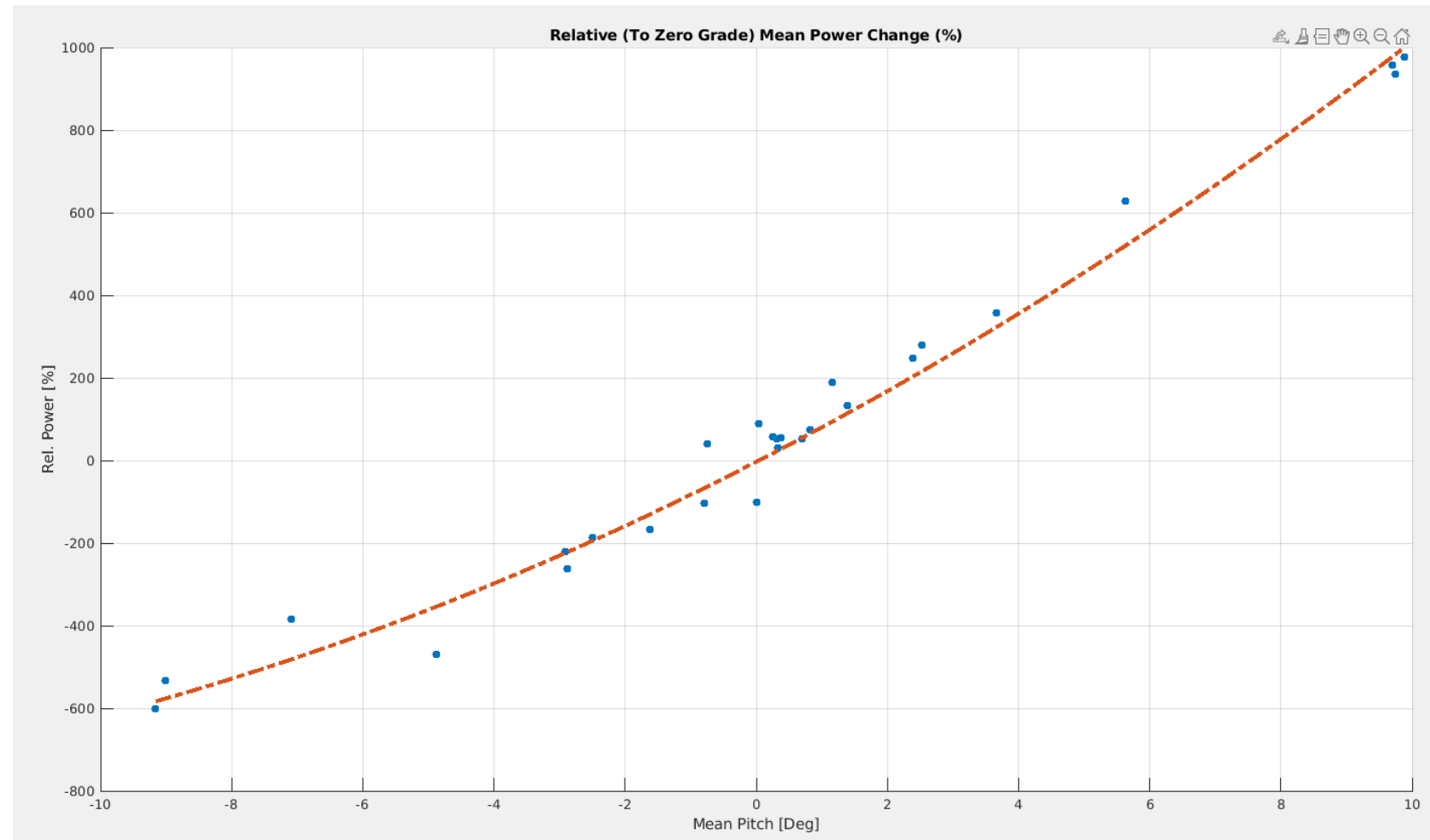
Data Analysis

- The mean Energy it takes to drive 1[Km] at a specific road slope
- The measurements seem to fit a parabolic model
- Going uphill and down hill is not the same (not bias)
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Data Analysis

- The relative change in power consumption
- Driving on a flat road is not the same as driving uphill and then downhill
- It is a dominant factor in planning for EV route planning



Discussion

Discussion

- How general the model should be (different types of cars, different configurations)?
- Will we be able to record some data for every car type to adapt the grade dependence model
- What's the accuracy we are looking for?
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- Next meeting
- Midterm report